



**SCIENTIFIC REPORT
OF
FOURTH INDIAN SCIENTIFIC EXPEDITION TO ANTARCTICA**

TECHNICAL PUBLICATION NO. 4

**DEPARTMENT OF OCEAN DEVELOPMENT
C G O COMPLEX, LODI ROAD
NEW DELHI 110 003
INDIA**

1987

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PREFACE

The Fourth Indian Scientific Expedition to Antarctica completed all its objectives, and the results of the scientific investigations largely carried out during the year 1984-85, have been presented in this report. The first wintering team which was left behind in Antarctica by the Third Expedition returned with the Fourth Expedition. Another wintering team consisting of thirteen members was left behind by the Fourth Expedition and all the members of this team also returned in the end of March 1986, after completing their task. The progress made by the successive Indian Expeditions to Antarctica, which started in 1981-82, has been steady and satisfying. By 1985, India has not only established a permanent station "Dakshin Gangotri", but also built another small summer station "Maitri" consisting of three cottages in the Schirmacher Oasis, a hanger and an alternate emergency accommodation near the permanent station. In addition, we have also established a High Frequency (HF) communication link between Dakshin Gangotri station and India.

The papers published on the Antarctic investigations by the Indian scientists during the last few years, now amounting to nearly 200, clearly indicate the growing interest in polar research in the country. At present, there are nearly 25 institutions in India working on the data, material and samples collected from Antarctica.

This Technical Report gives the details of the scientific work carried out by the Indian scientists in the disciplines of geology, geophysics, meteorology, biology, oceanography, polymer chemistry, renewable energy systems etc. It is greatly hoped that the Report of the Fourth Expedition will be as well received by the Antarctic scientists throughout the world, as the earlier three reports.

S.Z. QASIM
Secretary
Department of Ocean Development,
New Delhi

December, 1987

PARTICIPANTS OF THE SUMMER TEAM



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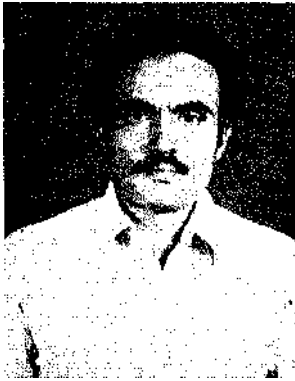
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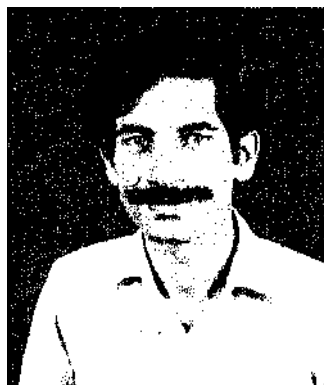
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A.K. Suri



S. Shivaji



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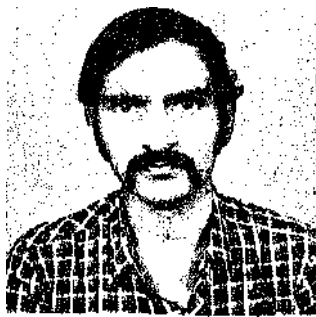
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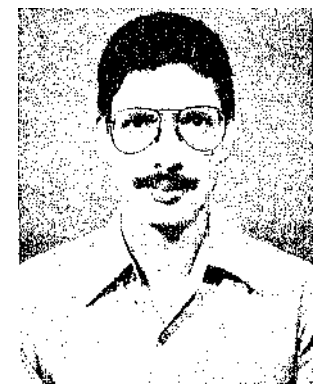
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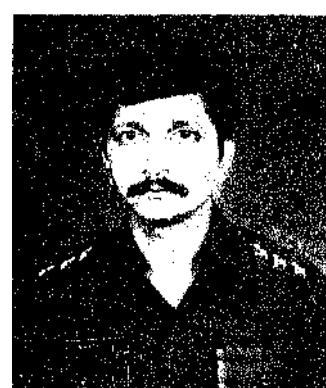
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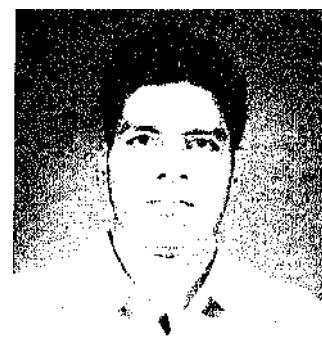
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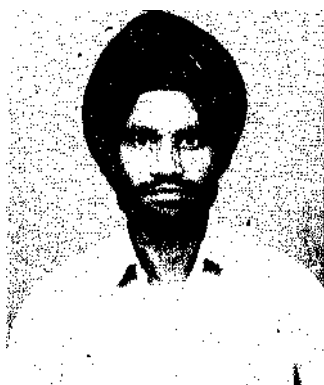
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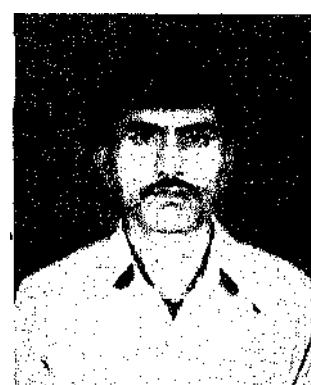
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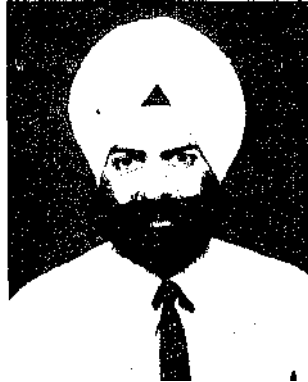
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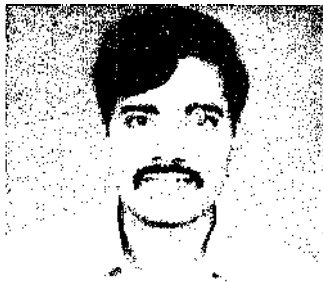
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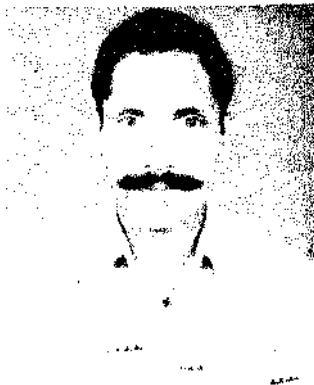
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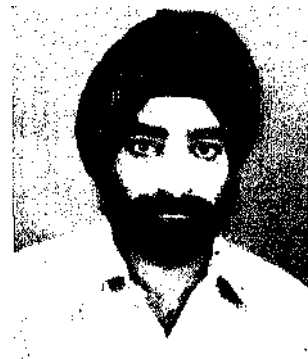
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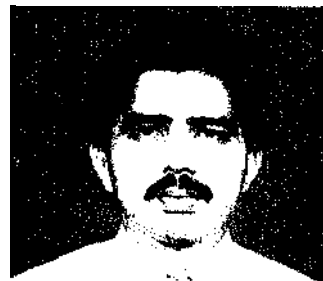
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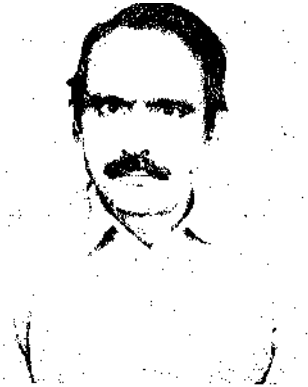
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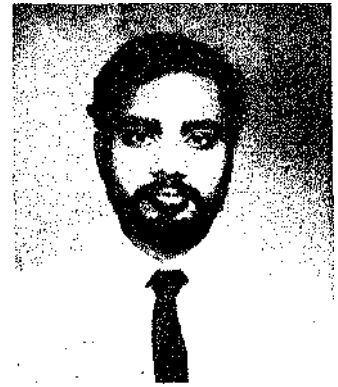
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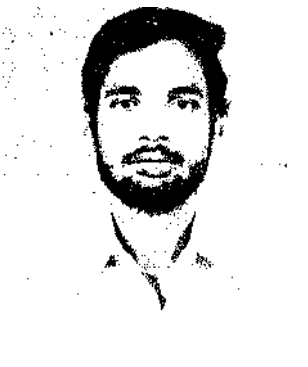
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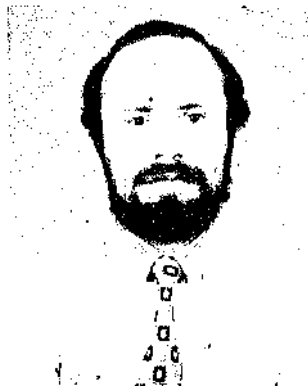
B.S. Dahiya



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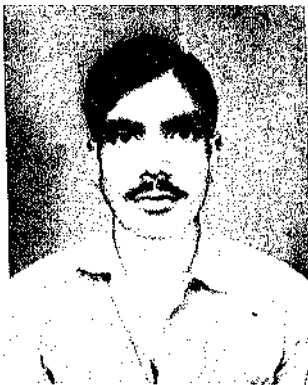
P. Aibara



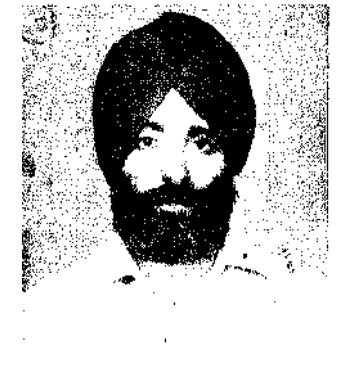
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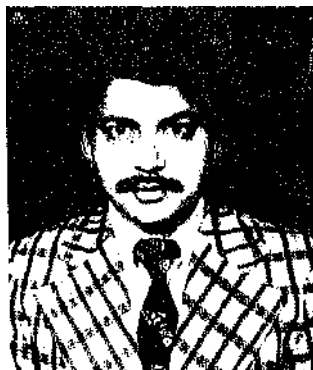
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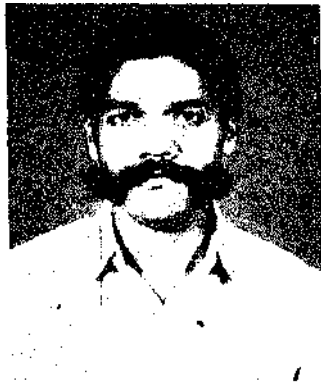
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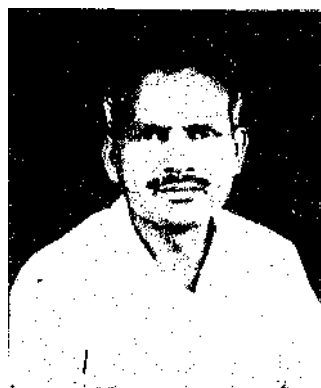
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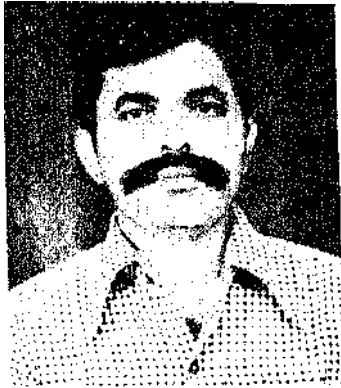
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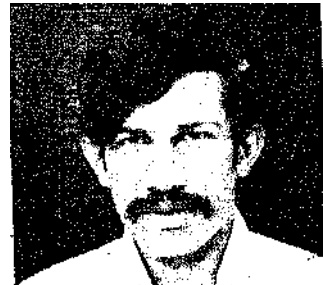
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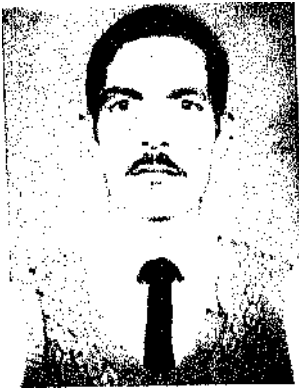
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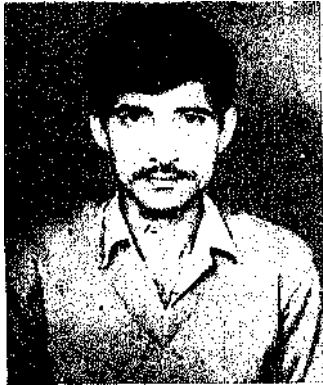
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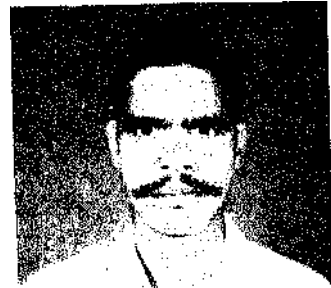
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INTRODUCTION

B.B. Bhattacharya

Department of Applied Geophysics, Indian School of Mines

The successful international cooperation in the course of the International Geophysical Year 1957-58 and the consequent international research programmes gave rise to growing interest of India in long term scientific work on the basic problems of the Antarctic environment. The activities date back to 1981 when the First Indian Expedition to Antarctica, led by Dr. S.Z. Qasim, hoisted the Indian National Flag on the ice shelf of Antarctica. Since then, the Indian Antarctic Expedition is an annual feature.

The Fourth Indian Expedition left Goa on December 4, 1984 with 82 members on a chartered ship *M.V. Finn polaris*. One scientist from Mauritius joined the team later at Mauritius. Till 65°S our ship did not face any obstruction either from floating or fast ice. But problems arose beyond it. Our progress was hampered by fast ice offering stiff resistance. The ship changed its course frequently but without any success. After 36 hours of gruelling effort on a particular route, on the morning of December 28, 1984, the ship ultimately entered the Polynya. The ship berthed temporarily in the afternoon of December 28, 1984 on the fast ice 60 km from the Antarctic coast.

The work of the expedition could be divided into two broad groups:

GROUP A

Logistical work for creating the infrastructure which included the setting up of H.F. communication link between Dakshin Gangotri (DG) station and India, construction of a garage (60 m X 10 m) with detachable covered 24 m long ramp and also with a facility of a modem workshop inside the garage for repairs and maintenance of the vehicles, construction of three cottages on a rocky terrain (oasis) about 75 km further south of the DG station and the construction of an alternate accommodation, 200 m away from the DG station.

GROUP B

Scientific investigations covering subjects like geophysics, geology, microbiology, meteorology, polymer chemistry etc.

To achieve these objectives the 83 members, coming from 18 different organizations including the three branches of the defence services and various scientific and academic institutions, participated in the work. The team had about two months time at its disposal out of which at least 15 days were not made available for any outdoor work due to bad weather, blizzards etc. Thus the actual days available were about 45—rather a limited duration—a fact which is not realised in many quarters. However, the team reached Antarctica during polar days, i.e., 24 hours of sunshine period which is a bonus as one could stretch working hours to much longer duration (14-16 hours).

It was clear to us that the single factor that would contribute most to the success of the expedition was the efficient handling and transportation of the cargo from the ship to the workspots. The task of the Fourth Expedition became more difficult as it had to work at two major sites separated by about 70-75 km. In addition, the team had more than 1000 tons of cargo to shift from the ship to the workspots and the permanent station. Considering all this it was decided that the cargo transportation should also be done by sledges driven by snow vehicles in addition to its transportation by heli-

copters. It was not an easy decision to take. So far, no Indian expedition had carried out a sledge operation on fast ice. Further many countries have met with accidents during sledging operations on fast ice and lost men and materials. These factors, however, did not deter us. After all in an expedition of this type one can not always play safe and still expect the expedition to succeed. Precautions, however, had to be taken to avert possible accidents. At the end of the expedition, it became quite obvious that the decision to operate sledges on fast ice for the transportation of cargo was a reasonable and sensible decision to achieve the goal, considering the amount of cargo and workload. The efficient handling of more than 1000 tons of cargo by the team could be considered a major success of the expedition.

The cargo had to be handled at three nodal points: (1) at the ship and (2) at two construction sites separated by a distance of about 70-75 km. Since the supply of the materials and their unloading at these sites had to be synchronised for a proper execution of the work, the logistic of the situation needed the unloading work at the ship to be assigned to a group of army and scientific personnel. This, though somewhat of an unpopular decision, had to be taken in the interest of the total objective. The problem was further aggravated as the ship could not berth at any place for more than six hours at a stretch during the entire period of the expedition—an experience not faced by the earlier Indian expeditions and generally not faced by the expeditions from other countries. The team unloaded the cargo from the ship at different places on different days and the coordination of the supply—a difficult job in itself, became more and more painstaking. The situation became somewhat like unloading of the materials on a certain day at a place X, the second at a place Y and the third at a place Z and so on, several hundred kilometres away. In the Antarctica expedition, however, one should be ready to face any eventuality.

In spite of all these problems, the team completed the work in time. The proper yardstick for the evaluation of the work should be to compare the accomplishments with those of the member countries of the "Antarctic Treaty" in their early stages. The work of the Fourth Expedition compares favourably with theirs, both qualitatively and quantitatively.

The logistical work mentioned under Group A is presented along with this introduction in the form of photocoverage (Figures 1-25) and also along with another paper on Schirmacher Oasis by the author. The scientific work on the various topics carried out by the scientific members of the team form the main contents of this report.

The Fourth Expedition completed all the tasks assigned to it and the aims and objectives were all fulfilled. The credit for these achievements, of course, goes to the team members. However, I would, as a leader of the expedition, like to single out the contributions of two of my Deputy Leaders, Lt. Col. P. Kumaresh of the Indian Army and Shri Dinesh Pandey, Geologist from the Oil and Natural Gas Commission.



Fig. 1. Glacier acclimatization training near Kargil, Jammu and Kashmir State (Photograph: B.B. Bhattacharya)



*Fig. 2. M.V. Finnpolaris:
The expedition ship
(Photograph: B.B. Bhattacharya)*



Fig. 3. Ship struggling to overcome the fast ice barrier (South of 65°S) (Photograph: M.P. Sinha)



Fig. 4. Planning of scientific work on board Finn polaris (Photograph: M.P. Sinha)



Fig. 5. Ship berthed to fast ice (Photograph: B.B. Bhattacharya)



Fig. 6. Reconnaissance survey of fast ice (Photograph: B.B. Bhattacharya)



Fig. 7. Major crevasse located in the fast ice adjacent to the shelf ice (Photograph: B.B. Bhattacharya)



Fig. 8. Snow accumulation around Dakshin Gangotri station after one year of its existence (Photograph: M.P. Sinha)



Fig. 9. Friendly Adélie penguins on fast ice (Photograph: M.P. Sinha)



Fig. 10. Adélie penguins shifting eggs during winter (Photograph: Lt. Cdr. Alope Banerjee)



Fig. 11. Unperturbed seal (Photograph: B.B. Bhattacharya)

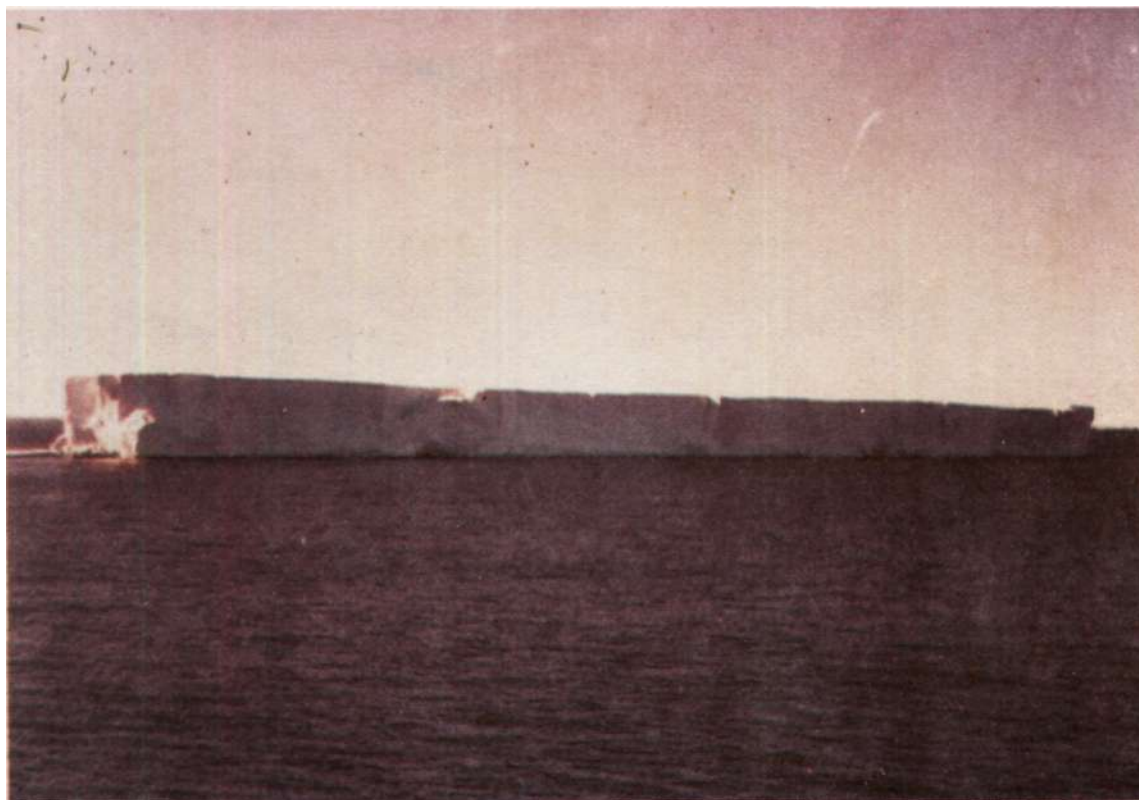


Fig. 12. Tabular iceberg (Photograph: B.B. Bhattacharya)

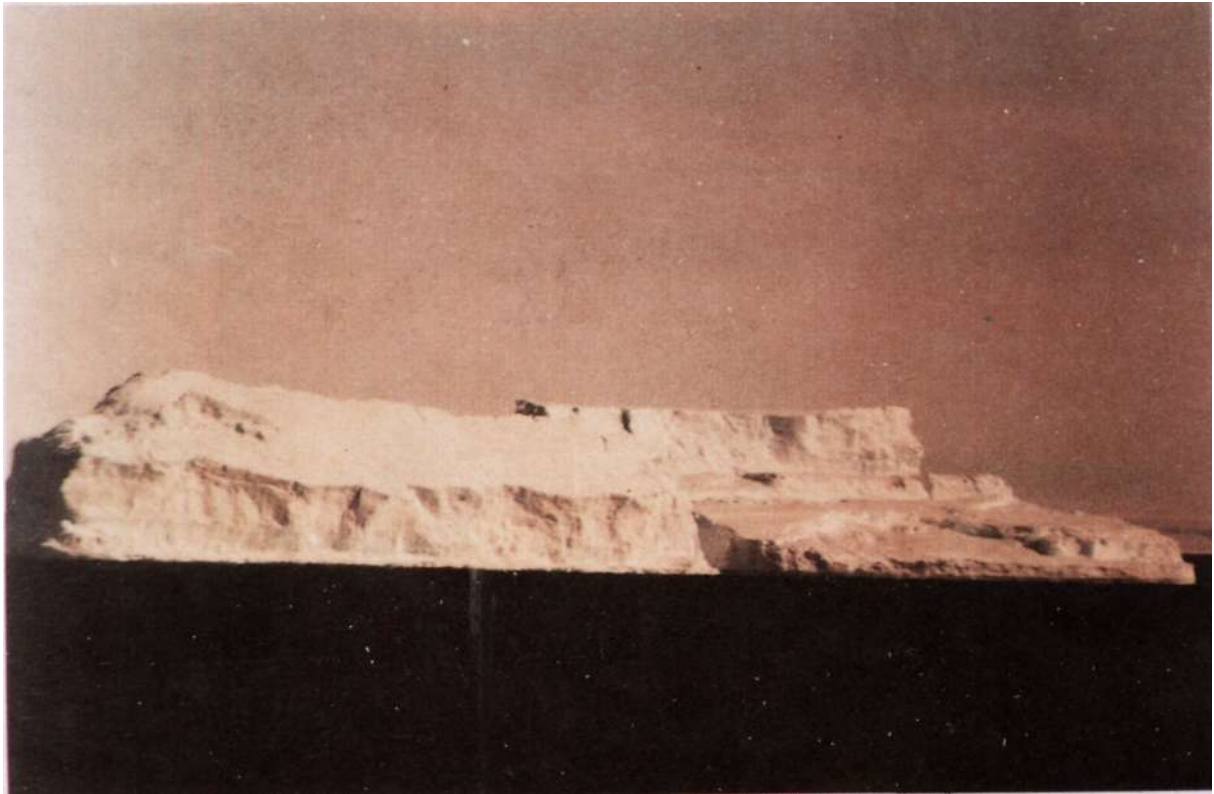


Fig. 13. Pyramid shaped iceberg (Photograph: B.B. Bhattacharya)



Fig. 14. Team members handling cargo (Photograph: B.B. Bhattacharya)



Fig. 15. Handling of large size cargo by the team members (Photograph: B.B. Bhattacharya)

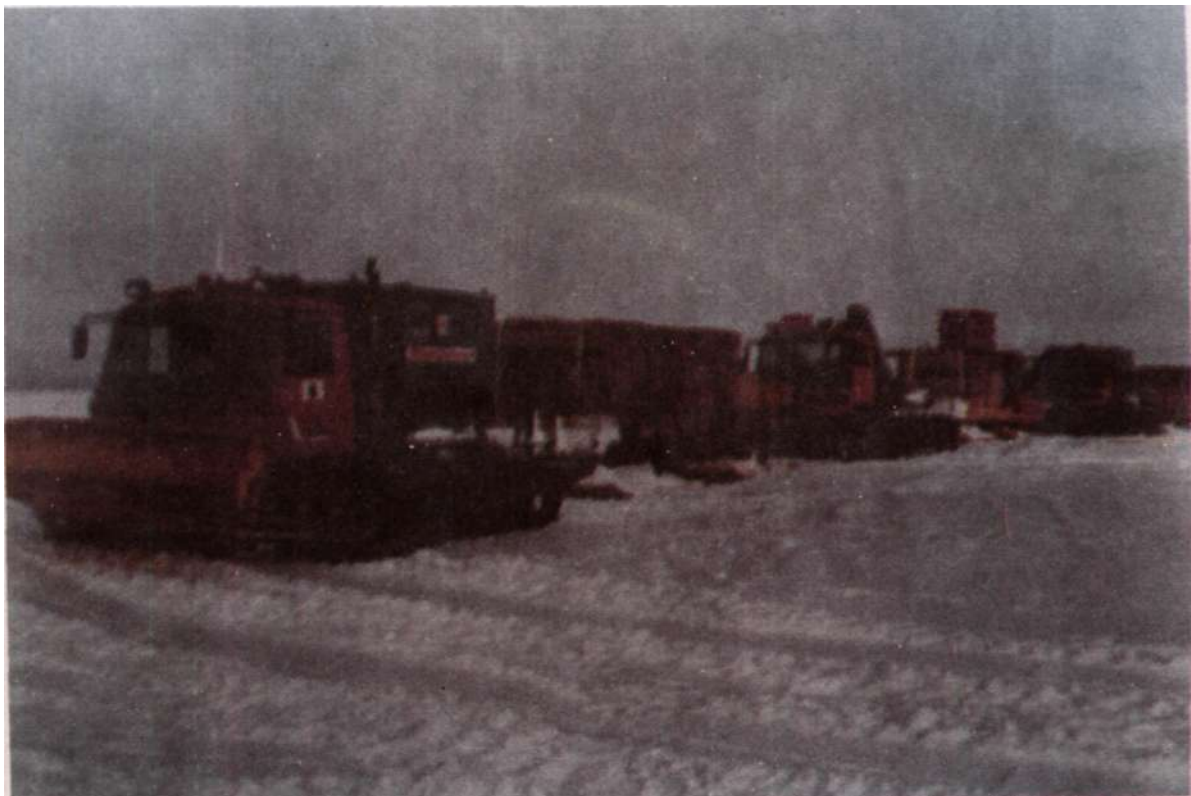


Fig. 16. Sledge operation over fast ice (Photograph: M.P. Sinha)



Fig. 17. Underslinging operation for the transport of cargo (Photograph: B.B. Bhattacharya)

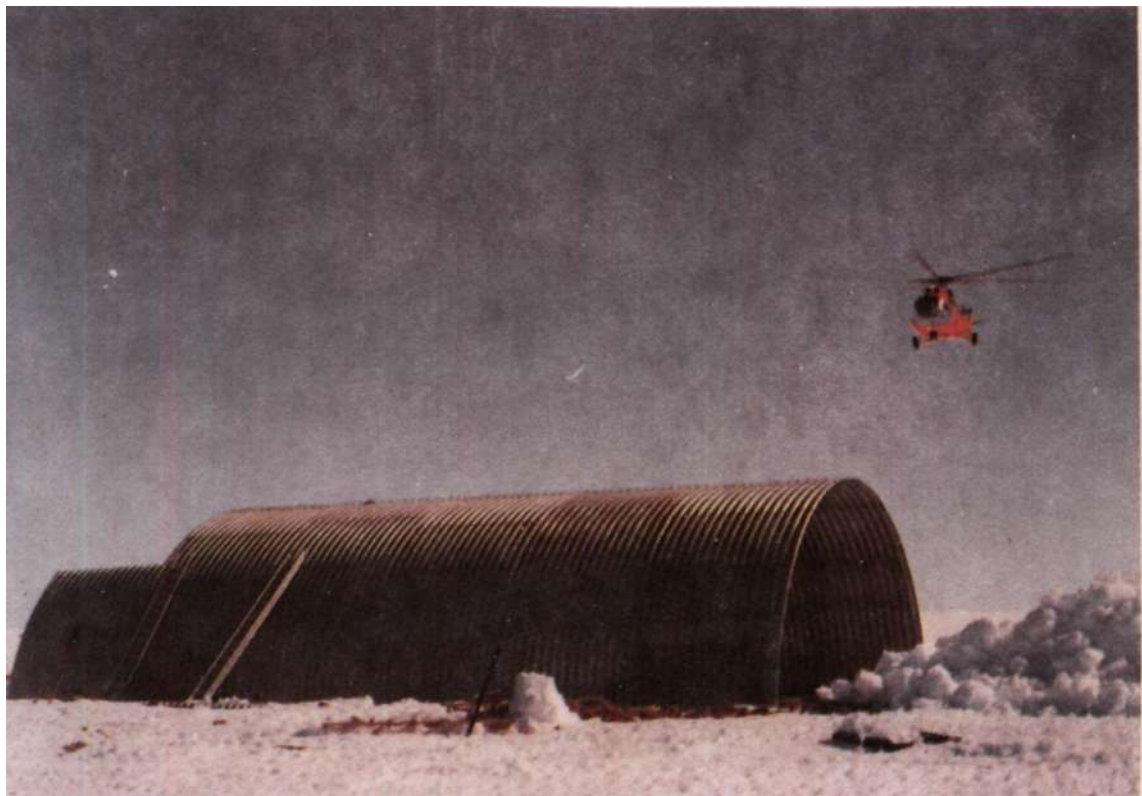


Fig. 18. Construction of garage near Dakshin Gangotri (Photograph: M.P. Sinha)



Fig. 19. Garage construction completed (Photograph: B.B. Bhattacharya)

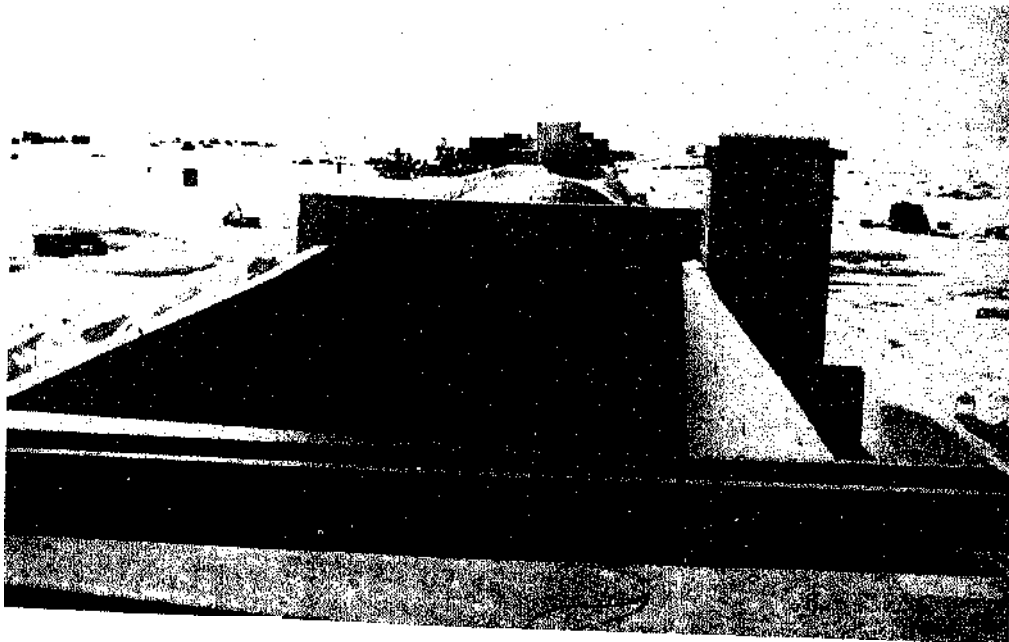


Fig. 20. Detachable ramp end of garage (Photograph; B.B.Bhattacharya)

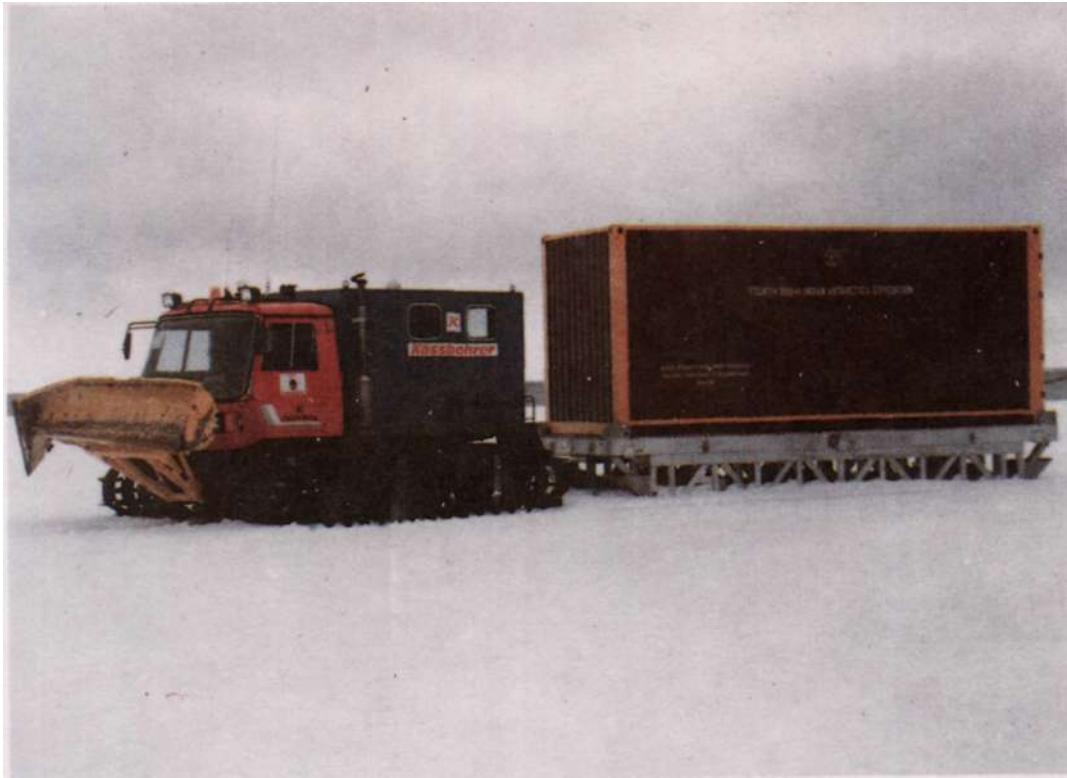


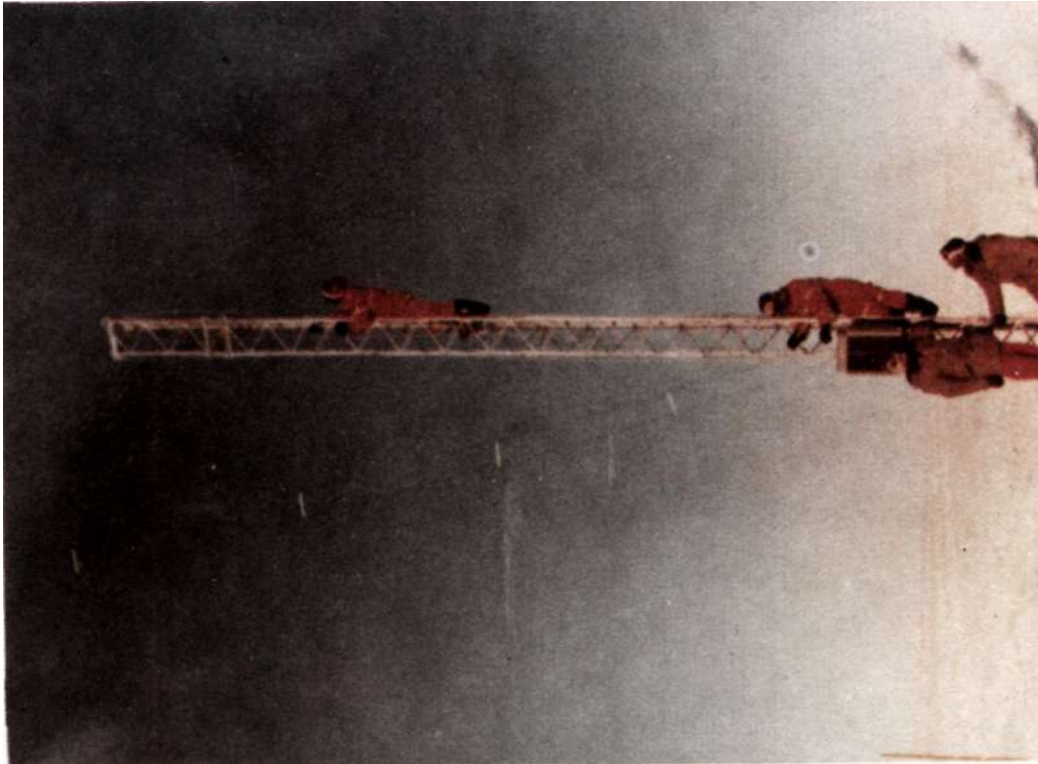
Fig. 21. Cargo for alternate accommodation near Dakshin Gangotri station being transported
(Photograph: B.B. Bhattacharya)



Fig. 22. Alternate accommodation in shelf ice near Dakshin Gangotri station
(Photograph: B.B. Bhattacharya)



*Fig. 23. Republic Day celebration by the Indian Team at
Dakshin Gangotri station (Photograph: M.P. Sinha)*



*Fig. 24. High frequency communication mast under erection near
Dakshin Gangotri station (Photograph: M.P. Sinha)*



Fig. 25. Sunrise in Antarctica in third week of February (Photograph: M.P. Sinha).